LINUX / SYSTEMS

SSH Visual Guide

Key auth, ssh-agent, tunnels, and hardening the daemon.

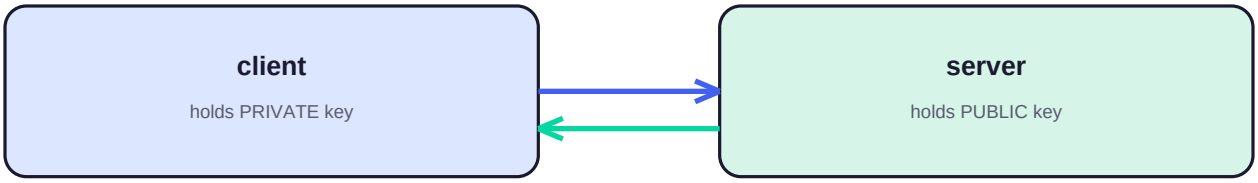
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Key-Based Auth: The Handshake

Prove identity with math, never send the secret.



1. client offers key id

2. server sends challenge

3. client signs with private key

4. server verifies → login

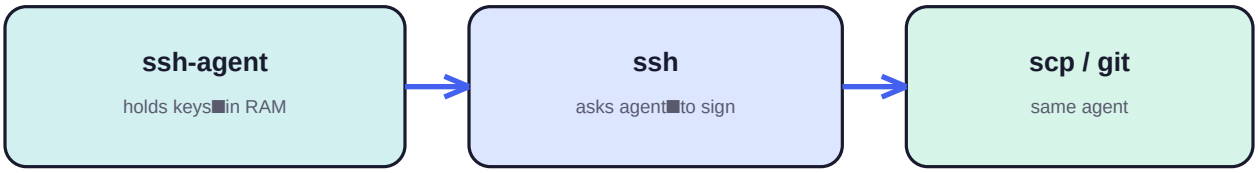
```
$ ssh-keygen -t ed25519 -C 'laptop'   # modern key type
$ ssh-copy-id user@host              # install public key
$ ssh -i ~/.ssh/id_ed25519 user@host
```

The private key NEVER leaves your machine. The server only ever sees signatures. Password auth is the fallback attackers brute-force — disable it (step 5).

STEP 2/6

ssh-agent: Unlock Once, Use All Day

The agent holds decrypted keys in memory; clients ask it to sign.




```
$ eval $(ssh-agent) && ssh-add ~/.ssh/id_ed25519
$ ssh-add -l            # keys currently loaded
$ ssh -A user@jump      # agent FORWARDING — convenient, risky
```

Agent forwarding (-A) lets the jump host use your keys — a compromised jump host can abuse them silently. Prefer ProxyJump instead: ssh -J jump@host target.

STEP 3/6

Tunnels: -L, -R, -D

SSH carries arbitrary TCP through the encrypted channel.

-L 8080:db:5432 (local forward)

localhost:8080 → server → db:5432

-R 9090:localhost:3000 (remote forward)

server:9090 → back through tunnel → your laptop:3000

-D 1080 (dynamic / SOCKS)

a local proxy; traffic exits at the server

```
$ ssh -L 8080:db.internal:5432 user@bastion    # reach private DB
$ ssh -R 2222:localhost:22 user@public       # expose laptop SSH outward
$ ssh -D 1080 user@host                     # SOCKS proxy on :1080
```

Remote forwards punch INBOUND holes — disable on servers you don't control (AllowTcpForwarding no) or attackers will pivot through you.

STEP 4/6

~/.ssh/config: Stop Typing Flags

Name your hosts once; every tool (ssh, scp, git, rsync) inherits it.

```
Host prod
  HostName 203.0.113.10
  User deploy
  IdentityFile ~/.ssh/prod_ed25519
  Port 2222

Host db
  HostName db.internal
  ProxyJump prod
  LocalForward 5433 localhost:5432
```

- ProxyJump chains bastions cleanly — no manual two-hop ssh, no agent forwarding needed.
- Match blocks apply settings per user/host: Match host db User readonly ...
- Keep secrets OUT of the config — IdentityFile points at keys, passphrases stay in the agent.

STEP 5/6

Hardening sshd

Defaults are convenient; production wants the strict set.

Setting	Secure value	Why
PermitRootLogin	no	root should never log in directly
PasswordAuthentication	no	keys only — kills brute force
PubkeyAuthentication	yes	the one allowed method
AllowUsers	deploy	everyone else rejected early
MaxAuthTries	3	slow down guessing
Port	2222*	obscurity only — do it WITH fail2ban

```
$ sudo apt install fail2ban        # bans IPs after N failures
$ sudo systemctl enable --now fail2ban
$ sudo sshd -T | grep -i permit   # verify EFFECTIVE config
```

Changing the port stops log noise, not attackers. Real defense = keys-only + fail2ban + updates. Test a NEW session before closing the old one — lockouts are forever.

STEP 6/6

Files Over SSH: scp, sftp, rsync

Same encryption, different ergonomics.

Tool	Best for	Example
scp	one-off copies	scp app.tar.gz prod:/opt/
sftp	interactive browsing	sftp user@host (like ftp, secure)
rsync -az	syncs, deploys	rsync -az --delete ./build/ prod:/srv/www/

- rsync -az: a=archive (perms+times), z=compress. --delete mirrors deletions — double-check the trailing slashes.
- Trailing slash matters: src/ copies CONTENTS; src copies the DIRECTORY. Get it wrong, get nested dirs.
- All three honor ~/.ssh/config — Host nicknames work everywhere.

STEP ✓/✓

Key Takeaways

SSH

- Key auth proves identity with signatures — the private key never crosses the wire.
- ssh-agent unlocks keys once; prefer ProxyJump over agent forwarding through bastions.
- -L forwards local→remote, -R remote→local, -D gives a SOCKS proxy; tunnels are TCP only.
- ~/.ssh/config names hosts once for ssh, scp, rsync, and git alike.
- Harden sshd: no root, keys only, AllowUsers, fail2ban — and test a new session before logging out.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev